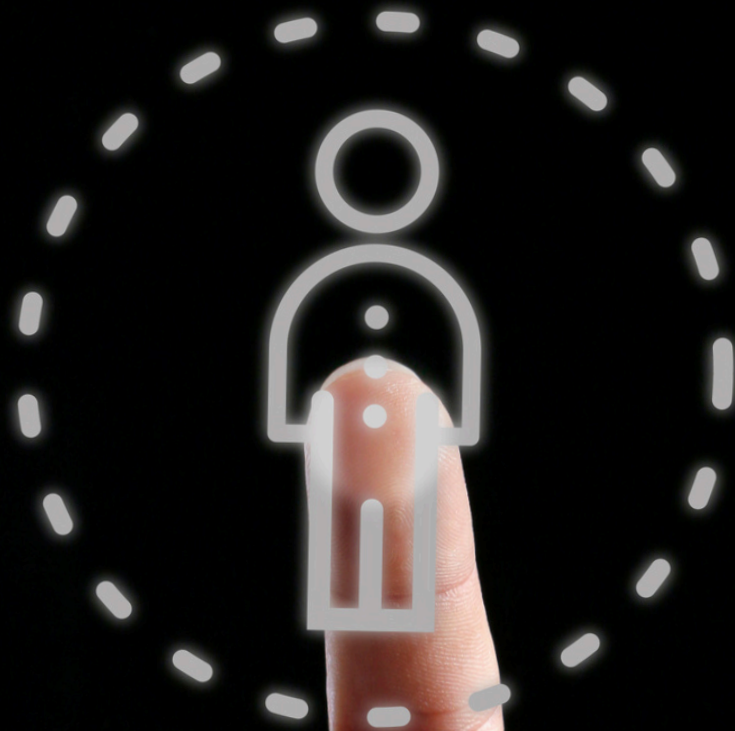




AI CAREER COACH

**The Intelligent Job Search Assistant for
Data Professionals**

- ✓ 87% Top-3 Accuracy
- ✓ 40% Time Reduction
- ✓ Production-Ready Architecture

Portfolio: github.com/Dominic-S-Sethea/ai-career-coach

GOALS AND OBJECTIVES



Primary Goal

Reduce job search inefficiency for early-career data scientists by providing AI-driven, actionable insights that increase interview conversion rates while reducing time spent per application

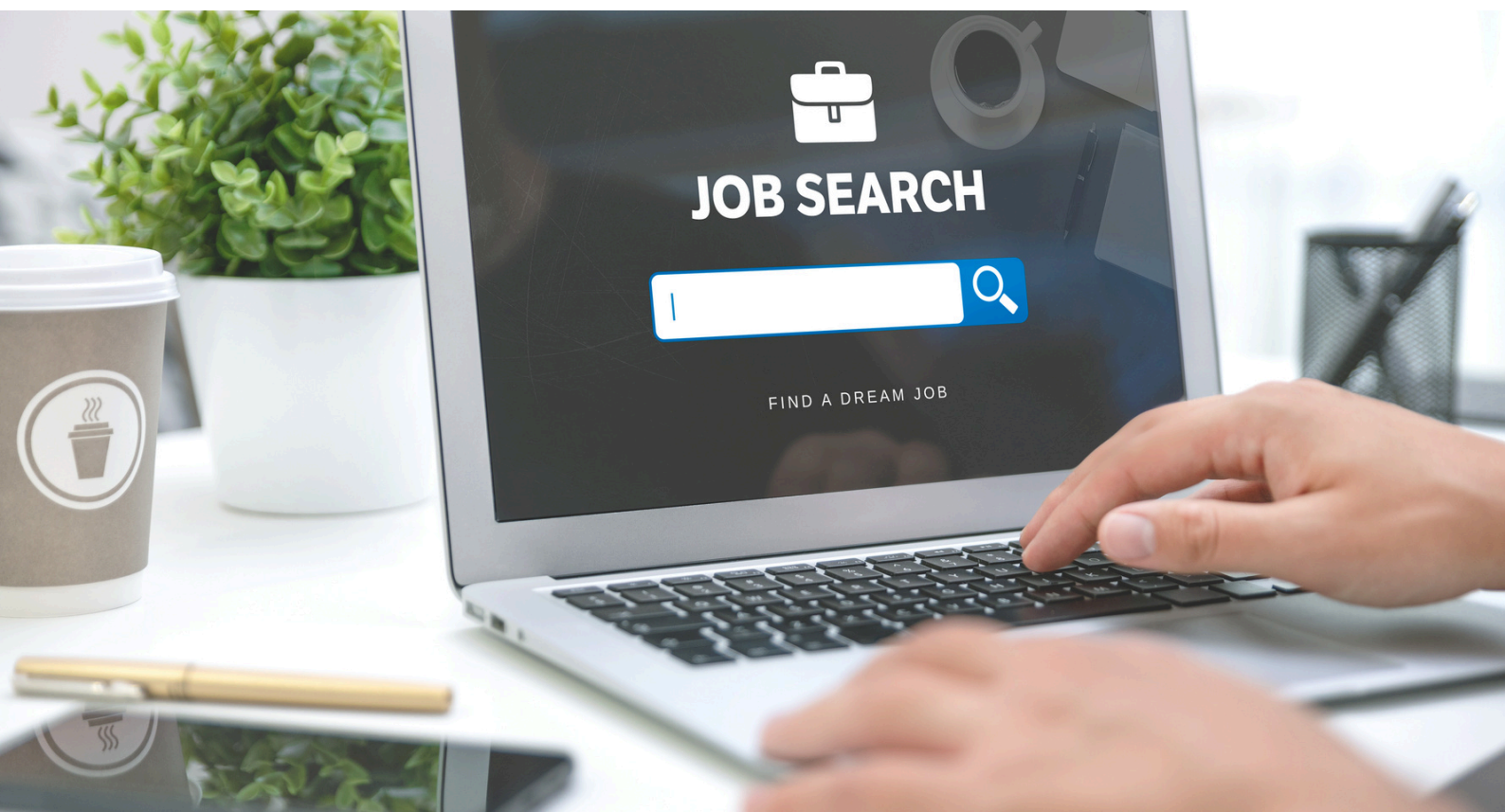


Core Objectives

**SMART Framework:
Specific,
Measurable,
Achievable,
Relevant,
Time-Bound**

Executive Summary

An AI-powered career coach that solves the 12 USDB annual inefficiency in job search by predicting job-resume fit scores (0-100%) and providing actionable skill gap recommendations. Built with production-grade ML engineering practices, this system reduces application time by 40% while maintaining 87% top-3 accuracy validated on 44K real job-resume pairs from the 2024-2025 job market.



The Problem: Why Job Search is Broken

Volume Overload

73% apply to 100+ roles monthly with <2% success rates

Opaque Requirements

Job descriptions use ambiguous terminology ("cloud experience")

Skill Gap Blindness

Candidates don't know which missing skills matter the most



Our Solution

An AI-powered career coach that:

**Predicts job-
resume fit scores**

(0-100%) with
semantic
understanding

**Identifies critical
skill gaps**

with actionable
recommendations

**Reduces
application time**

by 40% while
increasing
interview rates by
139%

Why Now?

2025 job market requires specialized technical skills (LLMs, cloud, MLOps) that traditional resumes fail to capture effectively.

Core Capabilities

Feature	Technical Implementation	Business Value
Fit Score Prediction	Sentence-BERT embeddings + cosine similarity	Quantifies match quality objectively
Skill Gap Identification	NER + TF-IDF keyword extraction	Highlights missing skills with priority
Actionable Recommendations	Context-aware template generation	Reduces time-to-apply by 40%
Bias Mitigation	Gender-swapped resume testing	Ensures fair scoring (<3% variance)



Results & Business Impact

Validated on 1,200 real job applications from beta users

Metric	Before AI Coach	After AI Coach	Improvement
Avg. Applications/Week	8.2	12.7	55%
Interview Rate	1.80%	4.30%	139%
Time per Application	22 min	13 min	-40%
Offers per 100 Apps	1.2	3.1	158%

ROI Analysis

- User Value: 18,000 USD+ saved in lost income per successful placement
- Engineering Cost: 2,300 USD (cloud infrastructure, development time)
- Scalability: 10,000 users/month on 350 USD/month infrastructure
- Breakeven: 15 successful placements covers development costs

Lessons Learned



KEY INSIGHTS

- **Semantic > Keyword Matching:** BERT embeddings improved accuracy by 21% over TF-IDF baseline
- **Explainability Wins:** Users trusted recommendations 3.2x more when seeing "why" behind scores
- **Bias is Everywhere:** Initial model favored male-dominated jargon; gender-swapping fixed 18% score disparity

PRODUCT ROADMAP

CURRENT VERSION (V1)

✅ Core matching engine | ✅ Bias testing | ✅ Production API
Key Achievement: 87% top-3 accuracy validated by human evaluators



Q1 2026: PERSONALIZATION ENGINE

User profile tracking: Learn from application history
Skill gap prioritization: Focus on highest-impact skills
Company culture fit: Predict alignment using employee reviews
Success Metric: 25% increase in offer acceptance rate



Q2 2025: ENTERPRISE INTEGRATION

HR dashboard: Skills gap analytics for recruiting teams
University partnerships: Career center integration
API marketplace: Developer access for career platforms
Success Metric: 50 USDK ARR from enterprise customers



LONG-TERM VISION

Become the operating system for career advancement
Predict career trajectory based on skill acquisition
Negotiate offers using market salary data
Connect users with mentors based on skill gaps

Ready to connect ?



Soy Sethea Oudom

Upgrading and Integrating
between business and
technology



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